DESCENT OF LOCAL NOETHERIANITY ALONG SURJECTIVE FLAT MORPHISMS LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one Properties of Spaces section and one Descent lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Local Noetherianity of algebraic spaces via \'etale charts. Stacks Project, Properties of Spaces, section spaces-properties-section-types-properties. Supplies that the property for X and Y is defined in terms of the charts U and V.
- fppf descent of local Noetherianity for schemes. Stacks Project, Descent, Lemma descent-lemma-Noetherian-local-fppf. Supplies the scheme-case conclusion.

Source reference. Original reference 03E5: sources/stacks-chow/tags.txt, source lines 4444–4444. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 034C: sources/stacks-chow/tags.txt, source lines 4087–4087. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of algebraic spaces. If f is locally of finite presentation, flat, and surjective and X is locally Noetherian, then Y is locally Noetherian.*

Proof. Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme U and a surjective étale morphism $U \to X \times_Y V$. As f is surjective, flat, and locally of finite presentation the morphism of schemes $U \to V$ is surjective, flat, and locally of finite presentation. In this way we reduce the problem to the case of schemes (as being locally Noetherian for X and Y is defined in terms of being locally Noetherian of U and V, see Properties of Spaces, Section 03E5). In the case of schemes the result follows from Descent, Lemma 034C. □

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